

Formation Engine Methodology

Final Statement — Master’s Thesis Conclusion

Formation Engine Development Team

Version 3.1.0

Contents

1	Restatement of the Thesis	2
2	Summary of Contributions	2
2.1	Scientific Problem Identification (§1)	2
2.2	Canonical State Representation (§0, §2)	2
2.3	Canonical Particle Container and Quantum Descriptors	3
2.4	Interaction Model (§3)	3
2.5	Thermodynamics and Integration (§4, §5)	3
2.6	Formation Physics and Statistical Classification (§6, §7)	4
2.7	Reaction and Electronic Extensions (§8, §8b, §9)	4
2.8	Polarization Models	4
2.9	Environment-Responsive Beads	4
2.10	Library Reactions and Thermodynamic Verification	5
2.11	BeadFIRE Visualization and Reporting Pipeline	5
2.12	Multiscale Projection and Future Architecture (§10, §12, §13)	5
2.13	Deterministic Pipeline and Self-Audit (§11)	6
3	Empirical Results	6
3.1	Formation Determinism	6
3.2	Variable Hierarchy Reduction	6
3.3	Causal Chain Confirmation	6
3.4	Validation Scale	7
4	What the Four Axioms Enabled	7
5	Declared Limitations	8
5.1	Coulomb Interactions Are Disabled	8
5.2	Orthogonal Boxes Only	8
5.3	Static Bond Graph	8
5.4	Coarse-Grained Layer Not Implemented	8
5.5	Force Field Accuracy	9

6	Forward Path	9
6.1	Immediate: Coulomb Stability Audit	9
6.2	Near-Term: Triclinic Cell Support	9
6.3	Near-Term: Visualization Frontend	9
6.4	Near-Term: SCF Polarization Integration	9
6.5	Mid-Term: Dynamic Bond Graph	10
6.6	Mid-Term: Atomistic to Coarse-Grained Mapping	10
6.7	Long-Term: ML Potential Correction	10
7	Situating the Work	10
8	Final Statement	10

1 Restatement of the Thesis

This work set out to answer a single, concrete question:

Can a deterministic, classical atomistic engine generate physically admissible molecular and crystalline structures from elemental identity alone, without reference to molecular databases, pre-solved crystal structures, or heuristic geometry templates — and can it do so in a manner that is fully reproducible, explicitly auditable, and scientifically credible?

The answer established over thirteen methodology sections, eleven validation suites comprising 1,013 tests, and an autonomous self-audit infrastructure is: **yes, within declared boundaries**.

The Formation Engine is that instrument. It is not a prediction claim. It is not a materials-discovery oracle. It is a reproducible computational tool for generating and evaluating candidate atomistic structures under explicit physical assumptions, targeted at the intermediate regime of 100–10,000 atoms where quantum methods become computationally intractable and continuum models must assume rather than derive structure.

2 Summary of Contributions

The contributions of this work are organized by layer.

2.1 Scientific Problem Identification (§1)

The foundational contribution is a precise diagnosis of the **emergence boundary** problem. The gap between quantum chemistry (limited to ~ 200 atoms at DFT level) and bulk continuum models is not merely a computational one. It is a *conceptual* one: existing frameworks have no natural language for structure as a computational output rather than as a starting assumption.

The formation engine resolves this by treating initialization not as a preparatory step but as a **primary experimental parameter**. Every structure generated by the engine carries full provenance: the random seed, the temperature and density boundary conditions, the force field parameters, and the convergence criteria. This converts initialization from an opaque procedural artifact into an auditable scientific record.

2.2 Canonical State Representation (§0, §2)

A complete, partition-respecting state representation was developed:

$$S = (N, \mathbf{M}, \boldsymbol{\tau}, \mathbf{X}, \mathbf{V}, \mathbf{Q}, \mathbf{B}, \mathbf{F}, \mathcal{E}, \mathcal{L}, \mathcal{B})$$

partitioned into three physically motivated tiers:

1. **Identity** ($N, \mathbf{M}, \boldsymbol{\tau}$) — immutable across the simulation lifetime.
2. **Phase** ($\mathbf{X}, \mathbf{V}, \mathbf{Q}, \mathbf{B}$) — governed by equations of motion; updated by the integrator.
3. **Scratch** ($\mathbf{F}, \mathcal{E}, \mathcal{L}, \mathcal{B}$) — fully determined by Phase; recomputed, never saved as independent state.

This partition is not cosmetic. It enforces a discipline that prevents common simulation errors: reading forces before they are evaluated, treating energy as an independent degree of freedom, or accumulating numerical drift in quantities that should be recomputed from scratch.

The per-particle identity vector $\mathbf{I}_i = (Z_i, A_i, Q_i, \Sigma_i, \Lambda_i, \Theta_i)$ grounds every interaction parameter in a single periodic table lookup, satisfying the **periodic table as sole authority** axiom.

2.3 Canonical Particle Container and Quantum Descriptors

The identity–state decomposition was extended in the unified methodology to a full canonical container representation:

$$\mathcal{C}_i = (\mathbf{I}_i, \mathbf{G}_i, \mathbf{C}_i^{\text{comp}})$$

where $\mathbf{G}_i = (s_i, c_i, f_i, \kappa_i, \eta_i)$ is the quantum descriptor bundle (spin, colour charge, flavour, projection coupling, information density) and $\mathbf{C}_i^{\text{comp}} = (\Gamma_i, \Xi_i, \Pi_i, \Upsilon_i)$ is the computational annotation payload carrying complexity class, kernel selector, solver priority, and annotation confidence.

The Container Invariance Principle states: $\mathcal{C}_i \equiv \mathcal{C}_j$ iff $\mathbf{I}_i = \mathbf{I}_j \wedge \mathbf{G}_i = \mathbf{G}_j$; computational layers may differ. This formalism ensures that particle identity is physics-determined, while computational treatment remains adaptive.

The framework was further extended to cell and world containers. A simulation cell is $\mathcal{K} = (\mathbf{K}^{\text{id}}, \mathbf{K}^{\text{geom}}, \mathbf{K}^{\text{comp}})$ with geometry defined by a 3×3 lattice matrix $\mathbf{L} = [\mathbf{a} \ \mathbf{b} \ \mathbf{c}]^\top$, $V = \det(\mathbf{L})$. A world (multi-cell simulation instance) is $\mathcal{W} = (W^{\text{id}}, W^{\text{top}}, W^{\text{ctrl}}, W^{\text{comp}})$ with a connectivity graph $G_{\mathcal{W}} = (V, E)$ between cells.

2.4 Interaction Model (§3)

The nonbonded potential is a Lennard-Jones 12-6 interaction:

$$U_{\text{LJ}}(r_{ij}) = 4\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \cdot S(r_{ij}; r_{\text{on}}, r_c)$$

with Lorentz-Berthelot combining rules ($\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}$, $\sigma_{ij} = (\sigma_i + \sigma_j)/2$) and a quintic switching function active over $[0.9 r_c, r_c]$ at $r_c = 10 \text{ \AA}$.

Bonded terms (harmonic bond, angle, torsion) are topology-aware and inferred directly from atomic positions via covalent radius matching with tolerance $f = 1.2$. This makes the bond graph a *derived* quantity of the geometry, not an independent input, consistent with the no-hidden-state requirement.

All parameters (ε , σ , r_{cov} , r_{vdW}) are drawn exclusively from UFF (Universal Force Field), covering $Z = 1\text{--}102$.

2.5 Thermodynamics and Integration (§4, §5)

The canonical ensemble is maintained via a velocity-rescaling thermostat, implemented as a *declarative* physical model rather than a numerical stabilizer. The thermostat’s effect on the energy budget is tracked explicitly in the energy ledger $\mathcal{E}$. The Nosé-Hoover chain extension (§4) provides a path to rigorously canonical sampling for future work.

Time integration uses the **Velocity Verlet** algorithm with default timestep $\Delta t = 1 \text{ fs}$. Numerical stability was verified through total-energy conservation tests with Coulomb disabled. The integrator produces bit-identical trajectories given identical initial state and seed, satisfying the **deterministic core** axiom.

Structure minimization uses the **FIRE** (Fast Inertial Relaxation Engine) algorithm, which was chosen for its compatibility with the existing Velocity Verlet integrator and its empirical convergence reliability on systems with 2–10,000 atoms. Convergence criteria are: $\max_i |\mathbf{F}_i| < 10^{-6} \text{ kcal}/(\text{mol } \text{\AA})$ and $|\Delta E|/N < 10^{-10} \text{ kcal/mol}$.

2.6 Formation Physics and Statistical Classification (§6, §7)

The formation pipeline transforms a seed and a molecular formula into a classified structure through a reproducible five-stage protocol: initialization, high-temperature exploration, staged annealing, FIRE minimization, and structural classification.

The classification system assigns every formation outcome to one of six categories: **stable**, **metastable**, **unstable**, **fragment**, **collapsed**, or **timeout**. Classification uses quantitative metrics — energy per atom, maximum residual force, bond-length deviation from reference, and coordination-number consistency — with explicit numerical thresholds, not heuristic labels.

Statistical analysis across seed ensembles (§7) provides formation probability, stability score, and structural diversity index. These are the quantities that make the engine useful as a *screening* tool: running 100 seeds for a given formula produces a statistically meaningful picture of its accessible configuration space.

2.7 Reaction and Electronic Extensions (§8, §8b, §9)

Section 8 documents the reaction engine: a graph-rewrite system that proposes, filters, and applies bond-topology changes under thermodynamic gating. Section 8b introduces heat-gated reaction control, in which reaction proposals are accepted or rejected based on local thermal energy rather than global temperature, enabling spatially heterogeneous reaction landscapes.

Section 9 documents the electronic state extension: partial charges via electronegativity equalization, polarizability corrections, and the interface to the (currently disabled) Coulomb term. The Coulomb implementation is structurally complete; re-enabling it requires only a dedicated stability audit of the integrator-Coulomb coupling.

2.8 Polarization Models

The unified methodology introduces self-consistent field (SCF) induced dipole polarization to address the 10–30% underestimation of hydrogen-bond energies and 50%+ underestimation for metal complexes inherent in fixed-charge models. Each particle acquires an induced dipole:

$$\boldsymbol{\mu}_i = \boldsymbol{\alpha}_i \cdot \mathbf{E}_i^{\text{loc}}$$

where the local field includes both permanent and induced contributions coupled through the dipole–dipole interaction tensor $\mathbf{T}_{ij}$. The resulting $3N \times 3N$ linear system $(\mathbf{I} - \mathbf{A})\boldsymbol{\mu} = \mathbf{E}^{\text{perm}}$ is solved iteratively. Thole damping with parameter $a \approx 2.6$ prevents the polarization catastrophe at short range. Atomic polarizabilities are drawn from Miller (1990) tabulations, maintaining the periodic-table-as-sole-authority axiom.

2.9 Environment-Responsive Beads

The coarse-grained layer was extended with environment-responsive beads that transform from static descriptor carriers into active material participants. Each bead carries an extended state:

$$\mathbb{B}_B = \underbrace{\{\mathbf{R}_B, \mathbf{Q}_B, \{c_{\ell m}^{(k)}\}, M_B, Q_B^{\text{tot}}, \mathbf{I}_B\}}_{\text{intrinsic}} \underbrace{\{\rho_B, C_B, P_{2,B}, \eta_B\}}_{\text{environment}}$$

Three fast observables (local density ρ_B , coordination number C_B , orientational order parameter $P_{2,B}$) are recomputed every dynamics step. A slow internal memory variable $\eta_B \in [0, 1]$ evolves via first-order relaxation:

$$\frac{d\eta_B}{dt} = \frac{1}{\tau} \left(\alpha \hat{\rho}_B + \beta \hat{P}_{2,B} - \eta_B \right)$$

where τ is the memory timescale and $\alpha + \beta = 1$. Environment state couples back into interaction kernels via $K_k(\ell, r; \eta_A, \eta_B) = K_k(\ell, r) \cdot (1 + \gamma_k \bar{\eta}_{AB})$, enabling emergent phenomena including spontaneous stacking, electrostatic screening, hysteresis, and nucleation asymmetry.

2.10 Library Reactions and Thermodynamic Verification

Three reference reactions were implemented with full Hess’s-law thermodynamics and mass-balance verification:

1. **Benzene nitration** (electrophilic aromatic substitution): $\Delta H_{\text{rxn}} = -35.56 \text{ kcal/mol}$.
2. **Thorium oxalate precipitation** (ligand exchange, product phase: sludge).
3. **Copper nitrate decomposition** (thermal, $\Delta H_{\text{rxn}} = +103.72 \text{ kcal/mol}$, endothermic).

Each reaction carries verified atom conservation through a mass-balance system that checks $\forall E : \sum_i \nu_i n_{E,i} = \sum_j \nu_j n_{E,j}$ with tolerance 10^{-6} .

2.11 BeadFIRE Visualization and Reporting Pipeline

A complete automated outcome-reporting and visualization pipeline was delivered for BeadFIRE (Lattice-Level FIRE) minimization. The pipeline comprises:

- C++ position recording at every FIRE step (stored in `BeadFIREStep::positions`)
- Structured Markdown report generation with convergence metrics, energy decomposition, and L^AT_EX equations
- Machine-readable CSV trajectory export with full bead position history
- Python visualization producing publication-quality (300 DPI) XY and XZ plane trajectory projections, energy convergence plots, force decay curves, and FIRE parameter evolution

All data flows from the actual FIRE kernel with no synthetic overlays, maintaining anti-black-box integrity. The system is 32-bit compatible and portable across platforms.

2.12 Multiscale Projection and Future Architecture (§10, §12, §13)

Section 10 establishes the four-tier scale hierarchy (molecular, cluster, bulk, coarse-grained) and defines the supercell construction protocol. Supercell geometry is replicated from the unit cell using:

$$\mathbf{r}_{i,pqr} = \mathbf{r}_i + p \mathbf{a} + q \mathbf{b} + r \mathbf{c}$$

with bond graphs re-inferred post-construction from atomic geometry. This ensures that topology is always consistent with the actual geometry, eliminating a common source of silent errors in periodic structure workflows.

Sections 12 and 13 document the extension architecture: the interfaces for machine-learning potential correction, reactive bond-order dynamics, triclinic cell support, and variable-cell relaxation. Each extension is specified at the interface level with explicit fallback requirements. The classical core remains independently evaluable regardless of which extensions are active.

2.13 Deterministic Pipeline and Self-Audit (§11)

The self-audit infrastructure is the enforcement mechanism for the reproducibility guarantee. It comprises three tools:

1. **Failure classifier** — autonomously categorizes outcomes from large run batches (>10,000 formations) into the six formation classes, producing structured JSON records.
2. **Gap targeter** — analyzes the coverage of parameter space and generates targeted new runs to fill identified gaps, enabling systematic exploration rather than random sampling.
3. **Regression detector** — compares classification outcomes across parameter versions, flags any molecule whose formation class or stability score changed, and enforces a binary verdict: NO REGRESSIONS OR REVIEW REQUIRED.

The reproducibility contract is strict: given (formula, seed, parameters), the engine produces **bit-identical output** across platforms, compiler versions, and run sequences. This is enforced by seed-controlled `std::mt19937` sampling, index-ordered force accumulation, and fixed-point summation for energy aggregation.

3 Empirical Results

3.1 Formation Determinism

The unified methodology establishes the Formation Determinism Postulate as a formal result:

$$M(t^*) = \mathcal{F}(\mathcal{E})$$

where the formation condition vector $\mathcal{E} = [T(t), c(t), p(t), P, R]$ captures thermal history, composition history, pressure history, pathway class, and environmental regime. Structure is a deterministic function of formation conditions — not an independent variable, not an assumption, but a computed output with full provenance.

3.2 Variable Hierarchy Reduction

Across the 1,013-test validation suite, the seven reported state variables $(\rho, C, P_2, \hat{\rho}, \hat{P}_2, f, \eta)$ were shown to reduce to four independent degrees of freedom:

$$\text{Independent: } \rho, C, P_2, \eta \quad \text{Derived: } \hat{\rho}, \hat{P}_2, f$$

This reduction simplifies the state space of the environment-responsive layer by nearly half, making parameter sweeps tractable and confirming that the derived quantities carry no independent information.

3.3 Causal Chain Confirmation

The empirically confirmed causal structure is:

$$\underbrace{\text{geometry}}_{\text{scene}} \longrightarrow \underbrace{\rho, C, P_2}_{\text{fast obs.}} \longrightarrow \underbrace{f}_{\text{target}} \longrightarrow \underbrace{\eta}_{\text{memory}}$$

Geometry determines fast observables. Fast observables determine the driving force f . The driving force determines the memory state η . No reverse causation was observed. The system exhibits pure first-order relaxation with no multistability.

3.4 Validation Scale

The validation infrastructure grew from the initial 35 Phase-1 audit checks to 1,013 tests across eleven suites:

Suite	Tests	Focus
Suite #1	60	Channel kernels, SH rotation
Suite #2	77	Observable correctness, η dynamics
Track 2	39	Monte Carlo robustness (4000+ trials)
Suite #3	52	Large- N formation studies
Suite #4	113	Dynamic regimes, invariance
Suite #5	131	Ensemble proxies, response validation
Suite #6	85	Macro property precursors
Suite #7	109	Property learning pipeline
Suite #8	193	Calibration, OOD detection
CLI Suite	72	Integration tests
Bead Layers	82	Visualization data model
Total	1,013	All passing

Key empirical findings across the suite:

1. Large- N runs ($N = 64, 125, 216$) are numerically stable.
2. Edge vs. bulk differentiation emerges naturally from the environment-responsive layer.
3. The causal chain geometry $\rightarrow$ observables $\rightarrow$ target $\rightarrow \eta$ is empirically confirmed without exception.
4. The system exhibits pure first-order relaxation with no multistability.
5. Monte Carlo robustness testing over 4,000+ trials confirms statistical stability.

4 What the Four Axioms Enabled

The formation engine was built on four design axioms (§1). In retrospect, each axiom had a specific payoff.

Explicit units everywhere. Eliminating reduced units forced every quantity to carry a physical meaning. This caught several parameter inconsistencies during development that would have been silent errors in a reduced-unit framework. It also makes outputs directly interpretable without a unit-conversion appendix.

No silent domain switching. Declaring the force field, integrator, and boundary conditions at session initialization made the simulation state globally inspectable at any point. There is no “which model is currently active?” ambiguity. This reduced debugging time significantly and made the regression detector straightforward to implement.

Deterministic core with stochastic exploration. Separating physics (deterministic) from exploration (stochastic, seed-controlled) made ensemble analysis scientifically meaningful. If two seeds produce different structures, that difference is *physical* (distinct metastable states), not numerical artifact. This distinction is essential for using the engine as a screening tool.

Periodic table as sole authority. The constraint to derive all parameters from Z alone eliminated an entire class of data-dependency problems. The engine can in principle evaluate any combination of elements up to $Z = 102$ without additional parameterization effort. This generality comes at the cost of accuracy for specific systems (UFF is not a high-fidelity force field), but that cost is *declared* and *bounded*.

5 Declared Limitations

Scientific honesty requires explicit enumeration of what this work does not accomplish.

5.1 Coulomb Interactions Are Disabled

The Coulomb term is implemented but disabled at the kernel level (`U_coul = 0.0`, `F_coul_r = 0.0` in `lj_coulomb.cpp`). This was triggered by an observed integrator-Coulomb coupling instability in NaCl dynamics. The root cause is identified as a timestep-dependent energy drift in the force accumulation path, but has not been resolved.

Consequence: all polar and ionic systems are currently evaluated using van der Waals interactions only. Relative energies for systems where electrostatics dominate (salts, charged biomolecules, strongly polar solvents) are not reliable. This does not affect noble gases, nonpolar organics, or metallic systems evaluated at the LJ level.

5.2 Orthogonal Boxes Only

Periodic boundary conditions are implemented for orthogonal boxes only. Non-cubic crystals (monoclinic, triclinic) require the full 3×3 lattice tensor $\mathbf{h}$ and its inverse. This extension is architecturally specified in §10 but not implemented.

5.3 Static Bond Graph

The bond graph $\mathbf{B}$ is inferred at initialization and does not update during dynamics. This is appropriate for rigid molecules and non-reactive materials, but it means bond-breaking and bond-formation events are not captured during production MD. The reactive dynamics extension (discrete graph-mutation rules and ReaxFF-style bond-order potentials) is architecturally specified in §13 but not implemented.

5.4 Coarse-Grained Layer Not Implemented

The xyzC format reserves slots for coarse-grained data (bead mapping, PMF reference, bulk density), but the atomistic-to-coarse-grained mapping is not implemented. All reserved slots are explicitly null in current output. They must never be interpreted as computed values.

5.5 Force Field Accuracy

UFF provides broad elemental coverage with moderate accuracy. It was not designed for high-precision energetics and is not competitive with system-specific force fields (OPLS, AMBER, CHARMM) for biomolecular or organic-chemical applications. The engine should be understood as a *structure generation* and *screening* instrument, not a quantitative thermodynamic predictor. Formations identified as stable candidates by the engine are appropriate inputs for refinement with higher-fidelity methods (DFT, CCSD(T)), not for direct comparison to experimental energies.

6 Forward Path

The following extensions are prioritized by their scientific impact relative to implementation effort.

6.1 Immediate: Coulomb Stability Audit

Re-enabling the Coulomb term is the single highest-impact near-term task. The required steps are:

1. Isolate the integrator-Coulomb coupling instability in a minimal NaCl dimer testcase.
2. Apply a standard fix (reduced timestep, Wolf summation truncation, or Ewald splitting) and confirm energy conservation in NVE dynamics.
3. Pass the Phase 2 Coulomb stability audit.
4. Uncomment the two disabled lines in `lj_coulomb.cpp`.

This unlocks the full intended domain of the engine.

6.2 Near-Term: Triclinic Cell Support

Replacing the orthogonal box representation with the full lattice tensor **h** enables simulation of all 14 Bravais lattices. This is required for the engine to function as a genuine crystalline structure generator. The extension is architecturally specified; implementation effort is estimated at one to two weeks.

6.3 Near-Term: Visualization Frontend

A dedicated atomistic visualization command (`viz ‘‘molecule name’’ --radius`) is the current active development direction. The BeadFIRE visualization pipeline (Markdown + CSV + Python figures) demonstrates the reporting pattern; extending it to the atomistic layer connects the engine’s outputs directly to an inspectable, figure-ready interface, closing the loop between computation and research reporting.

6.4 Near-Term: SCF Polarization Integration

The polarization model (SCF induced dipoles with Thole damping) is fully specified and parameterized. Integrating it into the production force loop re-enables accurate treatment of hydrogen-bonded and metal-coordination systems. The $3N \times 3N$ linear solve adds computational cost but is bounded by the existing cutoff infrastructure.

6.5 Mid-Term: Dynamic Bond Graph

Implementing discrete graph-mutation rules for bond formation and breaking converts the engine from a structure-relaxation tool into a genuine reaction-exploration tool. This is the enabling step for the heat-gated reaction-control system documented in §8b.

6.6 Mid-Term: Atomistic to Coarse-Grained Mapping

Implementing the CG mapping protocol populates the currently null slots in the xyzC format and enables the engine to serve as a bottom-up parameterization source for coarse-grained models, completing the intended cross-scale pipeline.

6.7 Long-Term: ML Potential Correction

The `MLCorrection` interface specified in §13 provides a structurally clean extension point for neural network potential corrections. The additive correction architecture $U_{\text{total}} = U_{\text{UFF}} + \Delta U_{\text{ML}}$ ensures that the classical core remains independently evaluable and that ML correction is strictly optional.

7 Situating the Work

The Formation Engine occupies a specific and intentionally narrow position in the computational materials science landscape.

It is **not** a replacement for DFT. Quantum accuracy requires quantum methods. The engine is intended to generate physically plausible candidates for DFT refinement, not to substitute for it.

It is **not** a machine-learning surrogate. It does not learn from data and does not generalize from a training set. It applies declared physics to declared inputs, making its domain and its failure modes explicit.

It is **not** a molecular dynamics package in the traditional sense. LAMMPS, GROMACS, and NAMD are equilibration engines optimized for production sampling. The Formation Engine is a *structure discovery* engine optimized for reproducible candidate generation.

What the engine *is*: a deterministic, auditable atomistic instrument for generating and classifying candidate structures in the intermediate size regime, grounded exclusively in the periodic table, and designed to serve as the first stage of a multi-fidelity computational investigation pipeline. With the unified methodology extensions — canonical particle containers, environment-responsive beads, SCF polarization, and automated reporting — the engine now spans from quantum descriptor bundles at the particle level to research-quality figure generation at the output level.

The anti-black-box design philosophy — every mapping decision, every parameter, every intermediate result inspectable and traceable — is not a stylistic preference. It is the scientific requirement that makes the engine’s outputs usable in a research context where results must be understood, not merely reproduced. The BeadFIRE visualization pipeline, which produces publication-quality XY and XZ plane projections from actual kernel trajectory data with no synthetic overlays, is a concrete expression of this philosophy: what the engine computes is what the researcher sees.

8 Final Statement

The work presented in this thesis establishes that a deterministic classical atomistic engine can generate, classify, and audit molecular and crystalline candidate structures starting from elemen-

tal identity alone, with no reference to experimental databases, heuristic templates, or opaque initialization procedures.

Thirteen methodology sections, eleven validation suites comprising 1,013 tests, three reference reactions with verified thermodynamics, an environment-responsive bead layer with empirically confirmed causal structure, a complete automated reporting and visualization pipeline, and an autonomous self-audit infrastructure constitute a functioning scientific instrument with fully declared scope, fully declared limitations, and a clear forward path.

The formation engine does not claim to solve materials discovery. It claims something more modest and more durable: **it makes the intermediate-scale structure generation step explicit, reproducible, and scientifically legible.**

The Formation Determinism Postulate — $M(t^*) = \mathcal{F}(\mathcal{E})$ — is the formal distillation of this claim. Structure is not assumed; it is computed. Formation conditions are not hidden; they are recorded. Results are not approximate; they are bit-identical.

In a field where initialization is routinely treated as a procedural detail, treating it as a primary experimental parameter is not a minor engineering choice. It is a statement about what computational science owes to reproducibility.

The engine exists to make that statement in working code.

Formation Engine v3.1.0 — 13 sections, 1,013 validation tests across 11 suites, Phase 1 audit passed, all tests passing.